



100 Days of Python

Day 36 — Practice Assignment: Lists, Complete Review

1. Given letters = ["a", "b", "c", "d", "e"], use slicing to print just ["b", "c", "d"], then use slicing again to print the list reversed.

2. Given nums = [10, 20, 30, 40, 50, 60], print every second item starting from index 0 using slicing (i.e. [10, 30, 50]), then print the last two items using negative indexing.

3. Given matrix = [[1, 2], [3, 4], [5, 6]], print the second row, then print just the number 4 by indexing into the nested list twice.

4. Given original = [1, 2, 3], create a copy called shallow using original.copy(), append 4 to shallow, then print both lists to show that original was not affected.

Hint: Assigning copy = original (without .copy()) would NOT create a separate list.

5. Given words = ["sun", "moon", "star"], use a list comprehension to build a new list called lengths containing the length of each word. Print lengths.

Hint: A list comprehension looks like [expression for item in list].

6. Given scores = [45, 88, 62, 73, 90, 55], use a list comprehension to build a new list called passed containing only the scores that are 60 or above. Print passed.

Hint: Add a condition: [item for item in list if condition].

7. Given inventory = ["pen", "pen", "eraser", "pen", "ruler"], write code that: (a) counts how many "pen" entries there are, (b) removes just one "pen", (c) inserts "marker" at index 1, and (d) finally slices out everything from index 1 onward. Print the result of each step.

Hint: Chain in order: count() -> remove() -> insert() -> slicing with [1:].

8. Given grid = [["x", "o", "x"], ["o", "x", "o"], ["x", "x", "o"]], flatten it into a single list called flat using a list comprehension with two for-clauses, then use .count() on flat to find how many "x" values there are in total.

Hint: A nested comprehension looks like [item for row in grid for item in row].